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Dynamic Ambulance Repositioning Using Offline Reinforcement Learning and Spatio-Temporal Graph Neural Networks

(Graduation Project Part 1) prepared to obtain the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

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Abstract

Emergency medical services must position a limited ambulance fleet so that patients can be reached quickly despite demand that varies across locations and time. This project studies dynamic repositioning: selecting a waiting station for an ambulance when it becomes available. The proposed pipeline forecasts hourly demand over a graph of H3 city cells using a Spatio-Temporal Graph Convolutional Network (STGCN), uses forecast-informed coverage scores from the Maximum Expected Covering Location Problem (MEXCLP), and evaluates Behaviour Cloning (BC) and Conservative Q-Learning (CQL) policies in an event-driven simulator.

San Francisco public dispatch records are replayed in the simulator. Because public repositioning decisions are unavailable, the offline dataset is generated by replaying historical training-period calls under specified behaviour policies; it is not a record of actual dispatcher repositioning choices. The current notebook reports 92.29% of calls reached within eight minutes for MEXCLP with STGCN and 92.28% for CQL anchored to MEXCLP on the 2025 hold-out period. The paired test reports no significant difference between them ($p = 0.961$). Thus, these results support a benefit from forecast-informed coverage in the simulator, but do not demonstrate that CQL improves on MEXCLP. All findings are simulation-based and require cautious interpretation.

Keywords: Emergency medical services; ambulance repositioning; STGCN; offline reinforcement learning; CQL; MEXCLP; discrete-event simulation.

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General Introduction

1.1 Background and Motivation

Emergency medical services (EMS) operate with limited ambulances while emergency demand changes over time and across neighbourhoods. The location of an available ambulance affects the time needed to reach the next patient. Dynamic repositioning changes the waiting location of an available ambulance after it completes a service, using current fleet status and expected demand.

1.2 Problem Statement and Research Question

Simple return-to-home or nearest-station rules are easy to apply but may ignore expected demand and the coverage provided by other ambulances. Coverage optimisation can use demand estimates, while reinforcement-learning methods can represent sequential decisions. Training an online policy by trial and error on a live medical service is unsuitable. This project therefore studies offline learning in a simulator, while explicitly recognizing that the available public dataset does not include real repositioning decisions.

Research question: Can an offline reinforcement-learning policy, provided with STGCN demand forecasts, improve ambulance repositioning over a forecast-informed MEXCLP policy when both are evaluated on the same simulated calls? The primary outcome is the percentage of calls reached within eight minutes.

1.3 Objectives, Scope and Contribution

Prepare ambulance-served medical-call data from San Francisco for chronological training, selection and hold-out evaluation.

Represent the mainland service area as H3 cells connected by travel-time-based graph edges.

Forecast next-hour demand and compare STGCN with historical-average and persistence baselines.

Replay calls in an event-driven simulator and compare static, nearest-station, MEXCLP, BC and CQL policies.

Measure forecast accuracy and operational outcomes, including response times and the share of calls reached within eight minutes.

Scope and limitation: this is a proof of concept on a simulator. The offline transitions are generated from historical calls under declared behaviour policies, not taken from agency repositioning logs. Therefore, results describe the simulated setting and do not establish real-world operational benefit.

Contribution: the project evaluates forecast-informed ambulance repositioning and offline policy learning under a common simulator and a chronological hold-out. It does not claim a new STGCN or CQL algorithm.

1.4 State of the Art

Ambulance repositioning research includes covering-location optimisation, simulation-based benchmarks, deep reinforcement learning and forecasting-informed allocation. Recent studies

show that the project's question is related to active work; the proposed distinction is specifically the evaluation of offline learning from simulator-generated fixed data against a forecast-informed optimisation baseline. These studies show that ambulance stationing, spatiotemporal forecasting and reinforcement learning are active research areas. The present project focuses on comparing simulator-generated offline learning with forecast-informed optimisation; this is a scoped distinction rather than a claim that no related combination exists.

Study	Approach and reported result	Relation and limitation
Strauß et al. (2022) [1]	Open, scalable San Francisco ambulance-redeployment simulator and benchmark comparison; reports that simple baselines can perform strongly.	Same city data and simulator context; supports fair baseline comparisons.
Sivagnanam et al. (2024) [2]	Hierarchical multi-agent RL with actor-critic agents, Transformer-XL actors and combinatorial allocation; Nashville and Seattle data; reports about three orders of magnitude faster decisions than MCTS and slightly lower response time.	Online proactive repositioning; differs from fixed-dataset offline learning.
Rautenstrauß and Schiffer (2025) [3]	Optimization-augmented learning for online dispatch and redeployment from full-information optimization solutions; San Francisco 2023 data; reports up to 30% lower mean response time and up to 87.9% runtime savings in study scenarios.	Uses the same city data, but a structured/imitation-learning pipeline rather than CQL/IQL.
Tsai et al. (2026) [4]	STGCN/DCRNN forecasting combined with PPO and a GCN allocation policy; considers fairness, coverage and cost; uses NormRMSE/NormEP and response-distance, coverage and fairness metrics.	Closest pipeline match; uses online PPO and different EMS datasets.
Kostrikov et al. (2022) [5]	Implicit Q-Learning uses expectile value regression and advantage-weighted behavioural cloning; evaluated on D4RL offline-RL benchmarks.	General offline-RL method and candidate for comparison; not an EMS application.

Foundational work remains relevant: MEXCLP provides a probabilistic coverage baseline; STGCN provides the graph-temporal forecasting architecture; Strauß et al. provide a real-data San Francisco redeployment benchmark. The selected studies span 2022–2026. Three are recent EMS applications (2024–2026); the 2022 and earlier works provide benchmark or algorithmic foundations. The review distinguishes recent EMS applications from foundational work on benchmarks and offline-learning algorithms.

1.5 Proposed Solution

The pipeline proceeds as follows: public dispatch and geographic data are cleaned and aligned; the city is represented by H3 cells and a road network; the data are split chronologically; STGCN predicts next-hour demand; a discrete-event simulator replays real calls and service durations; behaviour-policy replays create a fixed offline dataset; BC and CQL learn from those transitions; and all policies are compared on common hold-out days.

At a decision epoch, the agent observes fleet locations and status, time and workload features, demand forecasts and station-level MEXCLP gains. It selects an available station subject to capacity constraints. The simulator then measures realized response outcomes. The forecast

ablation compares no forecast, historical average and STGCN to isolate the forecasting contribution.

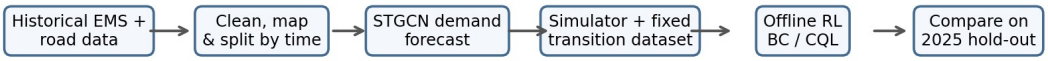


Figure 1. Proposed project pipeline. Offline transitions are generated by simulator replays under specified behaviour policies.

Chapter 1: Theoretical Study

1.6 EMS Performance Measures

Response time is modeled as queue waiting time plus turnout time plus emergency driving time. The primary measure is the percentage of calls reached within eight minutes. Secondary measures include mean, median and 90th-percentile response time, the percentage within ten minutes, queued-call percentage and decision latency. In the simulator these measures are estimates under its assumptions, not observed operational outcomes.

1.7 Covering Location Models

The Maximum Covering Location Problem seeks to place a fixed number of units to cover as much demand as possible within a time threshold. MEXCLP extends this idea by accounting for the probability that covering ambulances are busy. If an area has k covering units and unit busy probability q, its expected covered fraction is 1 − q^k. MEXCLP provides an interpretable forecast-informed baseline. In this project q is approximated from current system load, so the assumptions of independence and a common busy probability should be stated.

1.8 Spatial Graphs and STGCN

H3 partitions the city into hexagonal demand areas. A graph represents spatial relations between cells, with edge weights based on travel time. STGCN combines temporal convolutions with graph convolutions to model how demand changes over time and relates across nearby areas. The model predicts the next hour’s count per cell and is compared against historical-average and persistence baselines using Poisson deviance and complementary error measures.

1.9 Simulation and Decision Processes

A discrete-event simulator advances from call arrivals, ambulance status changes and station arrivals. Repositioning decisions occur when a vehicle becomes free, so the decision intervals vary. This is represented as a semi-Markov decision process (SMDP): state, action, reward, transition and elapsed time are specified. The simulator assumes a fixed fleet during an episode, one ambulance per call, modeled travel times and generated repositioning experience; these assumptions limit external validity.

1.10 Offline Reinforcement Learning

Offline reinforcement learning learns from a fixed dataset without collecting new actions during training. Behaviour Cloning imitates recorded actions. CQL adds a conservative value-learning objective intended to reduce overestimation for actions poorly supported by the dataset. Here “recorded” means transitions generated by simulator replays under known behaviour policies. The data-generation procedure and policies must be disclosed because conclusions depend on their coverage and quality.

1.11 Experimental Design

Chronological splitting reduces future-data leakage. The 2025 year is the final hold-out only if it is not used for training, tuning or choosing claims. The 2023–2024 test days were used during development/model selection in the current workflow, so they should be described as development or selection data, not an untouched final test. Paired comparisons across the same days and a Wilcoxon signed-rank test are reported. Forecast ablations assess the value of STGCN; fleet-size ablations assess sensitivity to system load.

Current result interpretation: on the available hold-out run, STGCN+MEXCLP reached 92.29% within eight minutes and anchored CQL reached 92.28%; the reported paired test found no significant difference ($p = 0.961$). The selected blend weight $w = 1$ corresponds to pure MEXCLP, so the result does not show a CQL contribution. These results do not demonstrate an improvement from CQL over MEXCLP.

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